Homework 3: Summarizing the "Why are Gaseous Halos Often Multiphase" Seminar

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Modeling the gas within the Circumgalactic Medium (CGM) is challenging because it is in a highly turbulent region that mixes multiphase gases. One of the biggest challenges of modeling the CGM is understanding the physics of how these gases mix with one another. This seminar aimed to explain the physics of why gaseous halos are seen to have multiphase gases and what are the main contributors to the mixing from a purely hydrodynamical standpoint. Dr. Fielding explains that the complex interplay of galactic winds, satellite galaxies, cosmic accretion, and thermal instabilities play a role in gas mixing. The physics of each of these contributors have been studied thoroughly for years and are at the point where they should all be utilized when modeling the CGM. By attempting to model this region with all of these hydrodynamic processes, the community could obtain a richer understanding of the CGM and the complex interplay of the gases found within them.

When looking at the CGM, it is important to distinguish the different types of gases found there. There is hot gas, whose temperature is approximately 10^6 K, cold gas at around 10^4 K, and an intermediate, or warm, gas at around 10^5 K.

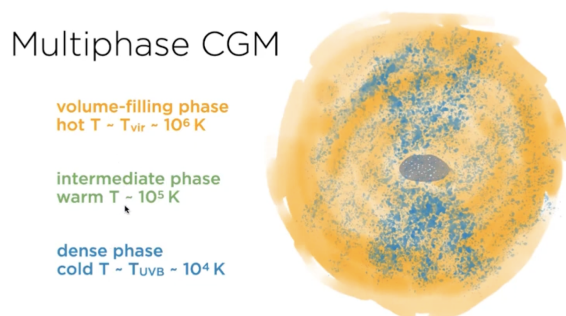


Figure 1: A screenshot of the multiphase CGM from Dr. Fielding's slides that displays the temperature scales of the gases.

The presence of all three different phases of gas already hint at the fascinating and complex physics, topics that have become of great interest within the field. Many studies have been published over the past decade trying to understand the individual dynamics that affect the mixing in the region. Dr. Fielding draws attention to how the cold gas phase evolves as it interacts with the other gases surrounding it in this seminar. Analyzing how the cold gas evolves over time can give us further insight on the different hydrodynamic physics that arise when multiphase mixing is present.

Galactic winds are one of the first discussed contributors to the multiphase gas mixing within the CGM, accelerating the cold gas to shear with hot gas. This shearing can contribute to turbulent mixing (like a Kevin Helmholtz instability) and create intermediate warm gas that can then cool down and fall into the cold gas, increasing the density of the cold phase and creating streams of cold gas. This growing mass can be represented by this equation:

$$\dot{M}_{grow} = \rho_{hot} r_{cloud}^2 \nu_{turb} (\frac{r_{cloud}}{t_{cool}\nu_{turb}})^{1/4}$$

However, to accelerate this mass of cold gas it must interact with the same amount of hot gas. So

$$M_{cold} = M_{hot}$$

in order to accelerate. Depending on the timescale, the cloud of cold gas can get destroyed while it's being accelerated but if the mixed gas, or intermediate gas, radiatively cools more rapidly than when the cloud of cold gas is dispersed, then the cold gas survives. So in order for your cold cloud of gas to survive,

$$t_{destroy} < t_{acceleration}$$

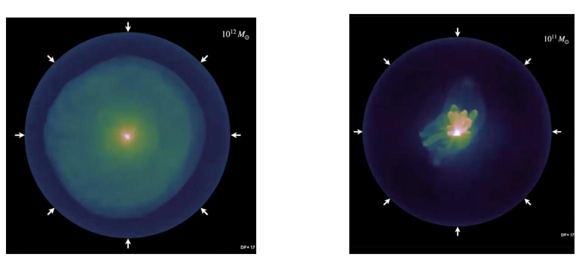
This indicates that as long as your acceleration time is larger than your destroy time, the mass of the cold phase is going to continuously grow, but this is not the case. The intrinsic turbulence between the two sheared gases can grow the cold gas, but the external turbulence, or background turbulence, of the hot gas will prevent the cold gas from growing. The mixing ideas laid out in this section of the seminar will be reused for the rest of the other mixing factors.

Satellite galaxies within the CGM can contribute to the mixing and fueling of cold gas due to their ram pressure stripping. Satellites can even generate their own winds. The stripping of gas can create tails of cool gas behind them, potentially allowing for the gas to cool faster than it can get dispersed. This allows for the same growing effect as we saw with the galactic winds.

Figure 2: A screenshot of Dr. Fielding's slides that shows satellite ram pressure stripping. The red circles indicate satellites where a clear tail is created behind them from the stripping

These tails can also cool down and create their own stars, complicating the dynamics even further. On the other hand, because satellite galaxies can create their own galactic winds, this adds to the turbulence and mixing of the multiphase gas.

Cosmic accretion from the Intergalactic Medium (IGM) can also contribute to multiphase mixing in the CGM due to the gas cooling post-shock and creating inflows. The accretion can happen when filaments enter into the CGM, creating turbulent shearing and allowing for the cold modes to grow, or IGM gases can enter in a smoother, more isotropic way. If the isotropic IGM gas cools rapidly, the mixing of the multiphase gas from the outer portion of the CGM is in relative thermal equilibrium, whereas if the gas cooled rapidly, there is a significant difference in how turbulent and thermally unstable the CGM becomes.



(a) Slow IGM Accretion into the CGM

(b) Fast IGM Accretion into the CGM

Figure 3: Screenshots taken from Dr. Fielding's slides which visualize the difference in turbulence between a slow and fast inflow of gas from the IGM.

The final major cause of multiphase mixing is due to the rapid cooling of the gas within the CGM. The above factors can cause rapid cooling which can lead to the gas accelerating at supersonic speeds. Once gas has reached supersonic speeds, you now have a thermal instability. To determine if a gas is thermally unstable, there are two time scales that we have to worry about: the free fall time and the cooling time. If the gas's cooling time is shorter than the free fall time, there is going to be a significant amount of thermal instabilities vs. when the cooling time is longer compared to the free fall time. These thermal instabilities only encourage more gas mixing, getting these gaseous halos to be multiphase.

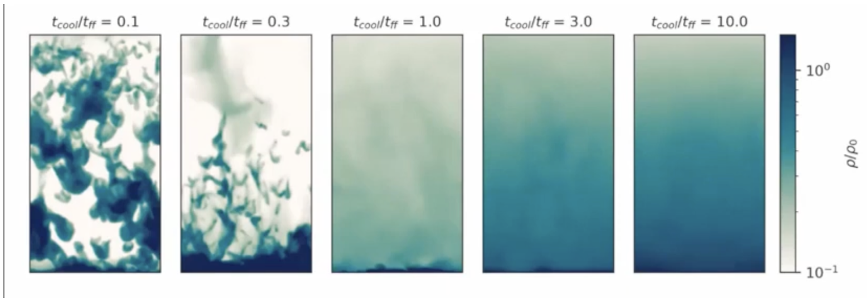


Figure 4: A screenshot from Dr. Fielding's slides that display the visual difference in mixing when looking at varying cooling vs. free fall timescales.

While each cause for the multiphase mixing has been discussed independently, it is important to study the cold gas mixing when several or all of these causes contribute. Hydrodynamic simulations are at the point where including all of these factors is possible, and we can get a richer and more thorough picture of the CGM when all of them are included. One question that I think would be interesting to study is how star formation and evolution within the cold gases add to the instabilities. It was briefly mentioned that the stripped gases from satellite galaxies have their own self gravity and can create stars, adding in more complexity to the hydrodynamics. Since the density of the cold phase gas can increase, this can lead to star formation. However, will the cold gas remain there long enough for stars to form, or will that cold gas become disturbed because of the surrounding environment? If stars do form, how does the radiation from the star play a role in the dynamics of the CGM environment? How does the surrounding area play a role in the evolution of the star? I believe this could be an interesting topic to explore while including all of the dynamics discussed in this seminar.